I must emphasize that these are just speculations based on the Medawar theory. Although there is a sense in which the Medawar theory logically must have some truth in it, this does not mean necessarily that it is the right explanation for any given practical example of senile decay. What matters for present purposes is that the gene selection view of evolution has no difficulty in accounting for the tendency of individuals to die when they get old. The assumption of individual mortality, which lay at the heart of our argument in this chapter, is justifiable within the framework of the theory.

The other assumption I have glossed over, that of the existence of sexual reproduction and crossing-over, is more difficult to justify. Crossing-over does not always have to happen. Male fruit-flies do not do it. There is a gene that has the effect of suppressing crossing-over in females as well. If we were to breed a population of flies in which this gene was universal, the *chromosome* in a 'chromosome pool' would become the basic indivisible unit of natural selection. In fact, if we followed our definition to its logical conclusion, a whole chromosome would have to be regarded as one 'gene'.

Then again, alternatives to sex do exist. Female greenflies can bear live, fatherless, female offspring, each one containing all the genes of its mother. (Incidentally, an embryo in her mother's 'womb' may have an even smaller embryo inside her own womb. So a greenfly female may give birth to a daughter and a grand-daughter simultaneously, both of them being equivalent to her own identical twins.) Many plants propagate vegetatively by sending out suckers. In this case we might prefer to speak of *growth* rather than of reproduction; but then, if you think about it, there is rather little distinction between growth and non-sexual reproduction anyway, since both occur by simple mitotic cell division. Sometimes the plants produced by vegetative reproduction become detached from the 'parent'. In other cases, for instance elm trees, the connecting suckers remain intact. In fact an entire elm wood might be regarded as a single individual.

So, the question is: If greenflies and elm trees don't do it, why do the rest of us go to such lengths to mix our genes up with somebody else's before we make a baby? It does seem an odd way to proceed. Why did sex, that bizarre perversion of straightforward replication, ever arise in the first place? What is the good of sex?*